

Ashish Kumar Singh

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SKILLS

• AI Architecture:	RAG Pipelines, Vector Databases, LLMs, Prompt Engineering, STT/TTS
• Machine Learning & MLOps:	Scikit-learn, Random Forest, PyTorch, SMOTE, SHAP
• Data & Visualization:	Streamlit, Power BI, PostgreSQL, MySQL, MongoDB
• Backend & APIs:	Fast API, REST APIs, JWT
• Tools:	Git, GitHub, Linux, shell scripting
• Languages:	Python, C++, SQL
• Relevant Coursework:	DSA, OOP, DBMS, Operating System, Computer Networks

PROJECTS

Offline Voice Assistant — Low-Latency STT-LLM-TTS Pipeline GitHub	Jun' 26
• Built a privacy-first, fully offline voice assistant by orchestrating a 5-stage pipeline (VAD → STT → LLM → TTS → Playback), achieving >92% end-to-end latency reduction (20s → 1.5s) through iterative model benchmarking • Optimized Speech-to-Text by migrating from Whisper to NVIDIA NeMo IndicConformer with CUDA-accelerated greedy decoding, achieving >11× speedup (8.5s → 0.75s) while supporting >9 Indian languages natively • Reduced LLM inference time-to-first-token by >86% (11s → 1.5s) through model profiling, parameter optimization (4B → 1B), and full GPU offloading; served async streaming responses using REST API integration • Engineered neural TTS synthesis with Kokoro-82M on PyTorch CUDA, delivering >5.7× faster audio generation (8.5s → 1.5s); built async audio pipeline using asyncio & aiohttp for non-blocking real-time playback • Tech: <i>Python, PyTorch, CUDA, NVIDIA NeMo, Ollama, Kokoro TTS, Silero VAD, Whisper, asyncio, aiohttp, NumPy, SciPy, HuggingFace, YAML</i>	
Production-Ready RAG System AI Retrieval Backend GitHub	May' 26
• Architected a production-grade hierarchical RAG pipeline over 4 Indian legal corpora using LangChain with Parent-Child chunking strategy, generating >4,900 child chunks and >1,200 parent chunks stored in Pinecone vector DB • Reduced query latency by 58% (12s to <5s) by optimizing Pinecone top_k (20 to 10), reducing query variants (4 to 2), and compressing Gemini embeddings (3,072 to 768 dimensions) • Designed an agentic workflow with dual-LLM routing — Groq (LLaMA 3.3 70B) for factual queries (<1s) and Gemini 2.5 Flash for complex multi-step reasoning; deployed via Docker on Render • Delivered 3 domain-specific legal NLP tools (Section Lookup, Citation Generator, Plain English Explainer) through FastAPI and Streamlit, indexed across ~5,000 legal vectors • Tech: <i>Python, FastAPI, Pinecone, Groq API, Gemini Embeddings, Docker, RAG</i>	

TRAINING

Lovely Professional University	Jun' 25 – Jul' 25
<i>From Data to Decisions</i> Link	
• Trained and benchmarked 3 ML models (Logistic Regression, Decision Tree, Random Forest) on 4,000+ patient records; performed model evaluation using precision, recall & F1-score achieving 84%+ accuracy • Built end-to-end data pipeline — EDA, feature engineering & preprocessing on 15+ clinical attributes using Pandas/NumPy; visualized insights via Power BI dashboards for stakeholder reporting • Tech: <i>MySQL, Python, Power BI, Machine Learning, Microsoft Excel</i>	

ACHIEVEMENTS

• Solved 240+ DSA problems on LeetCode (Trees, Graphs, Binary Search) Link • 2nd Runner-Up – Syntechs 2026 Hackathon (CodeChef) led 3-member team to build a production Agentic AI + RAG system with dual-LLM routing, FastAPI backend, vector search & real-time inference	
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CERTIFICATIONS

• Privacy and Security in Online Social Media Link	Oct' 25
• ChatGPT-4 Prompt Engineering & LLM Fundamentals Link	Aug' 25
• Build Generative AI Apps and Solutions with No-Code Tools Link	Aug' 25

EDUCATION

Lovely Professional University	Punjab, India
<i>Bachelor of Technology - Computer Science and Engineering; CGPA: 8.22</i>	<i>Since Apr' 23</i>
DAV Public School	Ranchi, Jharkhand
<i>Intermediate; Percentage: 82%</i>	<i>Apr' 22-Apr' 23</i>
Don Bosco Academy	Ranchi, Jharkhand
<i>Matriculation; Percentage: 94%</i>	<i>Apr' 20-Apr' 21</i>